

Aarin Mehta (He/Him)

aarinmehta.me | linkedin.com/in/mehtaaarin/ | (267)721-3457 | aarin1951@gmail.com

EDUCATION

University of Massachusetts Amherst

BS in Computer Science (GPA: 3.83)

Graduation Date: May 2026

- Awards: Dean's Honors List (All Semesters), Chancellor's Award

TECHNICAL SKILLS

- **Languages:** C++, Java, Python, JavaScript, SQL (PostgreSQL), Ruby, HTML, CSS
- **Cloud/Infra:** AWS (Lambda, Cognito, API Gateway, CloudWatch, X-Ray, DynamoDB, S3), Docker, Kubernetes
- **Tools/Frameworks:** OAuth 2.0, React, Node.js, FastAPI, gRPC/Protobuf, Git, RSpec, Jest, Jira, Agile

WORK EXPERIENCE

Amazon

New York, NY

Software Development Engineer

July 2026 - Present

- Leading **peak-readiness** for a **tier-1 authentication** service in the retail critical path during Black Friday / Cyber Monday, owning operational-readiness inputs across five categories (runbooks, mitigation levers, certificates, dependency risks, traffic forecasts) and validating capacity for a **~3x** surge to **~2.8M TPS** across **~10 production** regions.
- Extended a high-scale authentication service to issue placeholder transitive-auth (TA) tokens for unsupported identity providers, gating subject-less tokens behind a per-client feature flag to prevent unintended exposure, unblocking a downstream team stalled for **~1 month**
- Designed and built a distributed authorization cache for token issuance on a high-throughput authentication path, reducing p99 latency by **30%** and offloading repeated authorization checks from the critical request path

Amazon

New York, NY

Software Development Engineer Intern

May 2025 - August 2025

- Designed and developed an internal portal system, reducing network configuration lookup time by **70%** for cloud operations team
- Architected secure backend authentication using **AWS Cognito, API Gateway**, and internal services, improving access control validation by **40%**
- Achieved **99.9%** availability and reduced API response times by **60%** through serverless architecture using **AWS Lambda** and **API Gateway**
- Implemented comprehensive monitoring solution using **CloudWatch** and **X-Ray**, increasing incident detection speed by **45%**

UMass Amherst IT

Amherst, MA

Software Engineering Intern

December 2023 - April 2025

- Launched **6+** pages with custom routing for agile form submission, improving ease of use for **200** employees
- Enabled pinning of important messages by refactoring the public dispatch logs site, adopted by **50+** users, reducing message lookup time **30%**
- Optimized database queries and storage through custom controller actions for transportation data
- Implemented duplicate-trip feature for similar routes, saving analysts **7+** minutes daily

PROJECTS

FamilyCall - AI Wellness Companion Platform (<https://www.familycall.care>)

- Built a **C++** audio-processing module for decoding, resampling, and chunking call recordings ahead of transcription, optimizing a latency-critical stage of the pipeline
- Architected an event-driven backend for a production AI calling platform, where telephony events flow through a managed **message queue** to decoupled transcription and processing services, enabling independent scaling and fault isolation
- Built containerized microservices with a performance-critical service layer in **C++ over gRPC/Protobuf**, orchestrated with **Kubernetes**, backed by managed relational and object storage for structured data and call recordings
- Implemented **idempotent subscription billing** with retry logic and role-based access control for multi-member family dashboards, with **distributed tracing** and monitoring for pipeline latency and call-success tracking

Rideshare Web Application (<https://github.com/aarinmehta/RideshareUMass>)

- Engineered a scalable full-stack rideshare platform for the Five College community using React.js, Node.js, and PostgreSQL, enforcing secure access through a verified .edu **JWT-based authentication** system
- Architected an **automated ride-matching engine** that processes spatial and temporal constraints to pair users, successfully facilitating 150+ rides and reducing average coordination latency by 40% in a 50-user pilot
- Optimized **database performance** by implementing complex indexing and query refactoring in PostgreSQL, ensuring sub-second response times for ride searches and dashboard updates under concurrent load
- Ensured **system reliability and data integrity** by implementing robust transaction management to mitigate race conditions during simultaneous ride bookings, demonstrating a focus on building fault-tolerant, state-consistent systems